Additional Problems [for practice]

[1] Let p, q and r be propositions having truth values 0, 0 & 1 respectively. Find the truth values of the following

(i) $(p \wedge q) \vee r$ (ii) $p \rightarrow (q \rightarrow \neg r)$

(iii) $p \wedge \sim(r \rightarrow q)$ (iv) $\sim p \wedge (q \wedge \sim r)$

(v) $(p \wedge q) \rightarrow r$ (vi) $p \rightarrow (q \wedge r)$

[2] Let p and q be primitive statements for which the $p \rightarrow q$ is False. Determine the truth values of the following

(i) $p \wedge q$ — 0
(ii) $\sim p \vee q$ — 0
(iii) $q \rightarrow p$ — 1
(iv) $\sim q \rightarrow \sim p$ — 0

[3] Construct the truth table for the following

(i) $(p \wedge q) \rightarrow \sim r$

(ii) $p \rightarrow (q \rightarrow \neg r)$

(iii) $q \rightarrow \{(\neg p \vee r) \wedge \neg s\}$

(iv) $[(p \wedge q) \vee (\neg r)] \longleftrightarrow p$

(v) $q \leftrightarrow (\sim p \vee \sim q)$

(vi) $(p \rightarrow \sim q) \vee (q \rightarrow \sim p)$

(vii) $(p \veebar q) \rightarrow (p \wedge q)$

(viii) $(p \uparrow q) \uparrow q$

(ix) $(p \downarrow q) \uparrow \neg q$

(x) $(p \uparrow q) \wedge (p \downarrow q)$

[4] Prove that, for any proposition p, q, r, the following Compound propositions for Tautology

(i) $[p \to (q \to r)] \to [(p \to q) \to (p \to r)]$

(ii) $[\sim q \wedge (p \to q)] \to \sim p$

(iii) $[(p \to q) \wedge (q \to r)] \to (p \to r)$

(iv) $[(p \to q) \vee (p \to r)] \longleftrightarrow [p \to (q \vee r)]$

(v) $\{[(p \vee q) \to r] \wedge (\sim p)\} \to (q \to r)$

(vi) $[(p \vee q) \to r] \leftrightarrow [\sim r \to \sim (p \vee q)]$

[5] Prove the following:

Laws of Negation:

(i) $\neg(\neg p) \equiv p$

(ii) $\neg(p \wedge q) \equiv \neg p \vee \neg q$

(iii) $\neg(p \vee q) \equiv \neg p \wedge \neg q$

(iv) $\neg(p \to q) \equiv p \wedge \neg q$

(v) $\neg(p \leftrightarrow q) \equiv (p \wedge \neg q) \vee (q \wedge \sim p)$

$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$

[The Truth values in column 5 & 8 are identical

$\therefore$]

(vi) $p \to q \equiv \sim p \vee q$

(vii) $p \to (q \wedge r) \equiv (p \to q) \wedge (p \to r)$

(viii) $[(p \vee q) \to r] \Leftrightarrow [(p \to r) \wedge (q \to r)]$

(ix) $(p \leftrightarrow q) \wedge (q \leftrightarrow r) \wedge (r \leftrightarrow p) \Leftrightarrow [(p \to q) \wedge (q \to r) \wedge (r \to p)]$

(x) $p \to (q \to r) \Leftrightarrow p \to (\neg q \vee r) \Leftrightarrow (p \wedge q) \to r \Leftrightarrow (p \to q) \vee (q \to r)$

(xi)

[6] Write the Negations of the following.

(i) Two $\triangle$les are congruent Iff their Areas are Equal

(ii) If two Numbers are not Equal then their squares are not Equal

(iii) If a Number is real then it is either rational (OR) Irrational.

(iv) $p \wedge (q \vee r) \wedge (\neg p \vee \neg q \vee r)$

(v) $(p \wedge q) \rightarrow r$

(vi) $(p \vee q) \vee (\neg p \wedge \sim q \wedge r)$

(vii) $p \rightarrow (\neg q \wedge r)$

(viii) If Harold passes his C++ course and Finishes his data structures project then he will graduate at the End of the semester.

(ix) If x is not a real Number, then it is not a rational Number and Not an Irrational Number."

[7] Using the truth table, verify that each of the following is a logical implication.

(i) $[p \wedge ((p \wedge r) \rightarrow r)] \rightarrow (r \rightarrow s)$

(ii) $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$

(iii) $[(p \rightarrow q) \wedge \neg q] \rightarrow \neg p$

(iv) $[(p \vee q) \wedge \neg p] \rightarrow q$

(v) $[(p \rightarrow r) \wedge (q \rightarrow r)] \Rightarrow [(p \vee q) \rightarrow r]$

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(vi) $[p \wedge (p \rightarrow q) \wedge r] \Rightarrow [(p \vee q) \rightarrow r]$

(vii) $\{[p \vee (q \vee r)] \wedge \neg q\} \Rightarrow p \vee r.$

(viii) $(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r) \Rightarrow r$

8. Write the converse, inverse & contrapositive of the following

(i) $(p \wedge q) \longrightarrow (p \vee q)$

(ii) $p \rightarrow (\sim q \rightarrow \sim r)$

(iii) $[r \vee (\sim p \rightarrow q)] \longrightarrow \sim p$

(iv) $(p \longleftrightarrow q) \longrightarrow \sim r$

(v) If a quadrilateral is a parallelogram then its diagonals bisect Each other.

(vi) If a real Number x^2 is greater than zero then x is not Equal to zero

(vii) If a Δle is Not Isosceles then it is not ~~equ~~ Equilateral.

9. Write down the contrapositive of $[p \rightarrow (q \rightarrow r)]$ with

(i) Only one occurence of the connective $\rightarrow$

(ii) NO occurence of the connective $\rightarrow$

Sol: The Contrapositive of $[p \rightarrow (q \rightarrow r)]$ is given by

$$\neg(q \rightarrow r) \rightarrow \neg p.$$

(i) Consider $\neg(q \rightarrow r) \rightarrow \neg p \Leftrightarrow \neg\neg(q \rightarrow r) \vee \neg p$, $p \rightarrow q \equiv \neg p \vee q$

$\Leftrightarrow (q \rightarrow r) \vee \neg p$

$\quad$ only one $\rightarrow$

(ii) Consider $\neg(q \rightarrow r) \rightarrow \neg p \Leftrightarrow \neg\neg(q \rightarrow r) \vee \neg p$, $p \rightarrow q \equiv \neg p \vee q$

$\Leftrightarrow (q \rightarrow r) \vee \neg p$, Law of double Negation

$\Leftrightarrow (\neg q \vee r) \vee \neg p$, $p \rightarrow q \equiv \neg p \vee q$

$\quad$ No occurence of $\rightarrow$.

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